# Linear Algebra 2024 Final Solutions

## True or False Questions

Answer 'T(rue)' or 'F(alse)'. (You don't have to give a reason, but it might get you half a point if you are wrong.)

1. For any square matrix $A$, $\det(-A) = -\det(A)$. F: In general $\det(-A) = (-1)^n \det(A)$ where $n$ is the dimension of $A$.
2. If $B = M^{-1}AM$ then $\det(A^{-1}B) = 1$. T: $\det(A^{-1}B) = \det(A^{-1}M^{-1}AM) = \frac{\det(A)\det(M)}{\det(A)\det(M)} = 1$.
3. A matrix whose entries are 0s and 1s has determinant $1, 0$ or $-1$. F: $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} = -2$
4. If $x$ is an eigenvector of $A$ with eigenvalue $\lambda$ then it is an eigenvector of $A + aI$ with eigenvalue $\lambda + a$. T: $(A + aI)x = Ax + aIx = \lambda x + ax = (\lambda + a)x$.
5. The eigenvectors of $A$ are the same as the eigenvectors of $A^T$. F
6. For invertible $A$, the eigenvectors of $A$ are the same as the eigenvectors of $A^{-1}$. T
7. If $A$ and $B$ are diagonalisable then $AB = BA$ F: They have to be diagonalisable by the same matrix.
8. For any matrix $A$, the matrix $A^H A$ is Hermitian. T
9. All hermitian matrices are symmetric. F: **Real** Hermitian matrices are symmetric.
10. Every square matrix is similar to a triangular matrix. T: Shure

1. Find a third column so that

$$Q = \begin{bmatrix} \frac{1}{\sqrt{11}} & \frac{1}{\sqrt{6}} & \\ \frac{1}{\sqrt{11}} & \frac{2}{\sqrt{6}} & \\ \frac{-3}{\sqrt{11}} & \frac{1}{\sqrt{6}} & \end{bmatrix}$$

is orthogonal.

**Solution**

We pick $v = (1, 0, 0)$, which is clearly independent and remove its projections onto the other two columns:

$$\begin{aligned} v \quad - \quad & \frac{(1,1,-3)\cdot(1,0,0)}{11}(1,1,-3) - \frac{(1,2,1)\cdot(1,0,0)}{6}(1,2,1) \\ = \quad & \frac{1}{66}[(66,0,0) - 6(1,1,-3) - 11(1,2,1)] \\ = \quad & \frac{1}{66}(49,-28,7) = \frac{7}{66}(7,-4,1). \end{aligned}$$

Then we normalise to $\boxed{v = (7,-4,1))/\sqrt{66}.}$

2. Find a formula for the determinant of the $n \times n$ matrix $A_n = [a_{ij}]$ where $a_{ij} = 0$ if $i = j$ and $a_{ij} = 1$ otherwise:

$$A_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad A_3 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \qquad A_4 = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

.

**Solution**

It is $\boxed{\det(A_n) = (-1)^{n-1}(n-1)}$. You can verify this easily for $A_2$ and then show it by induction. For $A_n$, add $-1/(n-2)$ times a copy of every row but the first to the first row. This makes the first row $\begin{bmatrix} -\frac{n-1}{n-2} & 0 & 0 & 0 & \dots & 0 \end{bmatrix}$. Expanding along this row the determinant is

$$-\frac{n-1}{n-2}\det(A_{n-1}) = -\frac{n-1}{n-2}(-1)^{n-2}(n-2) = (-1)^{n-1}(n-1)$$

as claimed.
Another nice proof that someone came up with was to observe that the eigens were

$$\{(\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}, n-1), (\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, -1), (\begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \\ 0 \\ \vdots \end{bmatrix}, -1), \dots (\begin{bmatrix} -1 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}, -1)\}$$

and so the determinant, being the product of the eigenvalues, is $(n-1)(-1)^{n-1}$.

3. Show that for invertible $A$, the cofactor matrix $C$ has determinant $\det(A)^{n-1}$. (Hint: What is the cofactor equation for the inverse?)

**Solution**

(Note that the original problem on the test was asking you to prove something that was not true. I didn't remove marks if you made the mistake suggested by the original question. But you got extra marks if you noticed it was incorrect.) This is immediate by the cofactor equation for the inverse $A^{-1} = \frac{C^T}{\det(A)}$. Indeed $\det(C) = \det(C^T) = \det(\det(A)A^{-1}) = \det(A)^n \det(A^{-1}) = \det(A)^{n-1}$.

4. For the vectors $v = (3, 2)$ and $w = (1, 4)$, find the area of the triangle with sides $v, w$ and $v - w$.

**Solution**

This is half of the parallelopiped defined by $v$ and $w$ so has area

$$1/2 \left\| \begin{matrix} 3 & 1 \\ 2 & 4 \end{matrix} \right\| = 5.$$

5. Show that the sum of the eigenvalues of a matrix $A$ is the sum of its diagonal entries.

**Solution**

Expanding the characteristic polynomial $\prod_i^n (\lambda - \lambda_i)$ the co-efficient of $\lambda^{n-1}$ is $(-1)^n \sum_{i=1}^n \lambda_i$.
On the other hand, consider computing the characteristic polynomial as $\det(A - \lambda I)$ via the permutation formula. (The determinant is the sum, over all permuations of the columns, of the signed product of the diagonals. ) The only permuatitions of the columns of $A - \lambda I$ for which the product of the diagonals has degree at least $n - 1$ in $\lambda$ is the identity permutation. So the co-efficient of $\lambda^{n-1}$ of $\det(A - \lambda I)$ comes from this factor $\prod_{i=1}^n (a_{ii} - \lambda)$. The co-efficient of $\lambda^{n-1}$ in this is $-\sum_{i=1}^n a_{ii}$, as needed.

6. Compute $A^{100}$ where $A = \left[ \begin{smallmatrix} 4 & 3 \\ -2 & 11 \end{smallmatrix} \right]$

**Solution**

Using the standard characteristic polynomial method you can find the eigenvalues $\lambda_1 = 5$ and $\lambda_2 = 10$ and the correspoding eigenvectors $v_1 = (3, 1)$ and $v_2 = (1, 2)$. With this we diagonalise

$$\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}^{-1},$$

and compute

$$\begin{aligned} A^{100} &= \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 5^{100} & 0 \\ 0 & 10^{100} \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 5^{100} & 0 \\ 0 & 10^{100} \end{bmatrix} \frac{1}{5} \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix} \end{aligned}$$

7. In an epidemic, the populations $w_t, s_t$ and $d_t$ of 'well', 'sick' and 'dead' at time $t$ are described by the Markov process

$$\begin{bmatrix} d_{t+1} \\ s_{t+1} \\ w_{t+1} \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{4} & 0 \\ 0 & \frac{3}{4} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} d_t \\ s_t \\ w_t \end{bmatrix}.$$

   (a) How many of the sick people die in a give time step?

   (b) What is the steady state of the process?

**Solution**

In a time step 1/4 of the sick people die. The system has only the one eigenvalue 1 of absolute value 1 (the others are 3/4 and 1/2) and its eigenvector is $(1, 0, 0)$. So the steady state is that everyone is dead!

8. Solve the differential eqution $u' = Au$ where

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 6 \\ 0 & 0 & 4 \end{bmatrix} \qquad \text{and} \qquad u(0) = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

**Solution**

$A$ has eigenvalues $\lambda_1 = 1, \lambda_2 = 3$, and $\lambda_3 = 4$ with corresponding eigenvectors $x_1 = (1,0,0)$, $x_2 = (1,1,0)$, and $x_3 = (13/3, 6, 1)$. (The first is easy to see, for $x_2$ and $x_3$ you might have to do some small calculations.) So

$$A = \begin{bmatrix} 1 & 1 & 13/3 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 3 & \\ & & 4 \end{bmatrix} \begin{bmatrix} 1 & -1 & 5/3 \\ 0 & 1 & -6 \\ 0 & 0 & 1 \end{bmatrix}.$$

(Again computing the inverse of the conjugating matrix might have taken some work.) We get then that

$$\begin{aligned} u(t) &= \begin{bmatrix} 1 & 1 & 13/3 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} e^t & & \\ & e^{3t} & \\ & & e^{4t} \end{bmatrix} \begin{bmatrix} 1 & -1 & 5/3 \\ 0 & 1 & -6 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \\ &= \frac{8}{3} e^t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - 6e^{3t} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + e^{4t} \begin{bmatrix} 13/3 \\ 6 \\ 1 \end{bmatrix} \end{aligned}$$

9. The $3 \times 3$ matrix $A$ has two independent eigenvectors and has eigenvalues

$$\lambda_1 = 2 \qquad \lambda_2 = \lambda_3 = 3$$

What is its Jordan canonical form of $A$?

**Solution**

As it has two eigenvectors instead of three, the eigenspace for the repeated eigenvalue 3 had dimension 1 so there is only one Jordon block for this eigenvalue:

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix}.$$

10. (a) What is the Hermitian $A^H$ of the matrix

$$A = \begin{bmatrix} i & 4 \\ -1 & 2+i \end{bmatrix}?$$

(b) What is a Hermitian matrix?

(c) What is a Unitary matrix?

(d) Show that if $A$ can be diagonalised $A = U\Lambda U^{-1}$ by a unitary matrix $U$, then $AA^H = A^H A$.

Solution

$$A^H = \begin{bmatrix} -i & -1 \\ 4 & 2-i \end{bmatrix}?$$

A matrix $A$ is Hermitian if $A = A^H$. It is Unitary if $AA^H = I = A^H A$. So for unitary matrixes $U$ we have that $U^H = U^{-1}$. Thus if $A = U\Lambda U^{-1}$ for unitary $U$ we have that

$$\begin{aligned} AA^H &= U\Lambda U^H (U\Lambda U^H)^H \\ &= U\Lambda U^H U\Lambda^H U^H \\ &= U\Lambda\Lambda^H U^H \\ &= U\Lambda^H \Lambda U^H \\ &= U\Lambda^H U^H U\Lambda U^H \\ &= A^H A. \end{aligned}$$

Where $\Lambda$ and $\Lambda^H$ commute because they are diagonal.